AC9M6N07 • Year 6 Mathematics

Fractions, Decimals and Percentages of Quantities

Find familiar parts, discounts and remaining amounts efficiently

We are learning to…

Students connect equivalent fractions, decimals and percentages, find benchmark parts by division and multiplication and distinguish discount amount from sale price.

Represent | Reason | Calculate | Interpret | Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

solve problems that require finding a familiar fraction, decimal or percentage of a quantity, including percentage discounts, choosing efficient calculation strategies and using digital tools where appropriate

Find 25%, 40% and 75% of $240

part	strategy	amount
25% = 1/4	**240 ÷ 4**	**$60**
40% = 4/10	**240 ÷ 10 × 4**	$96
75% = 3/4	240 ÷ 4 × 3	$180

Use a familiar equivalent form that makes the quantity easy to partition. A 25% discount is $60; the sale price is $180.

Build discounts and percentage increase in context

20% discount on \$85	10% of 360	0.5 of 74	3/5 of 90	digital tool
discount \$17; pay \$68	36	37	54	compare many prices after formula is set

Always identify whether the question asks for the part, the remaining amount or the new total.

Build and annotate the model

Represent fractions, decimals and percentages of quantities and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Discount amount reported as final price:** Subtract the discount from the original.
- **Percent converted to whole number multiplier:** 20% = 0.20, not 20.
- **Quantity divided by percentage number only:** Use a fraction, decimal or benchmark strategy.
- **Whole not identified:** State the original total before finding a part.

Quick check

- Find 25% of 240.
- Find 3/5 of 90.
- Calculate a 20% discount.
- Distinguish discount/sale price.
- Choose a benchmark strategy.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.